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Course Code: ITA0305

Course Name: Mobile Computing

Experiment 6: Effect of OFDMA Downlink Scheduler on User Throughput and Fairness

Program Code:

```
import numpy as np

import matplotlib.pyplot as plt

# User channel conditions

users = 4

channel = np.array([10, 20, 30, 40])

# Round Robin Scheduler

rr = np.ones(users) * 25

# Proportional Fair Scheduler

pf = channel / np.sum(channel) * 100

# Calculate Jain's Fairness Index

def fairness(x):

    return (np.sum(x)**2) / (users * np.sum(x**2))

# Display output

print("----- OFDMA Scheduling Results -----")

print("User    RR Throughput    PF Throughput")

for i in range(users):

    print(f"User {i+1}      {rr[i]:.2f} Mbps       {pf[i]:.2f} Mbps")

print("\nRound Robin Fairness:", round(fairness(rr), 2))

print("Proportional Fairness:", round(fairness(pf), 2))

print("\nResult: User throughput and fairness calculated successfully.")

# Plot graph

x = np.arange(users)

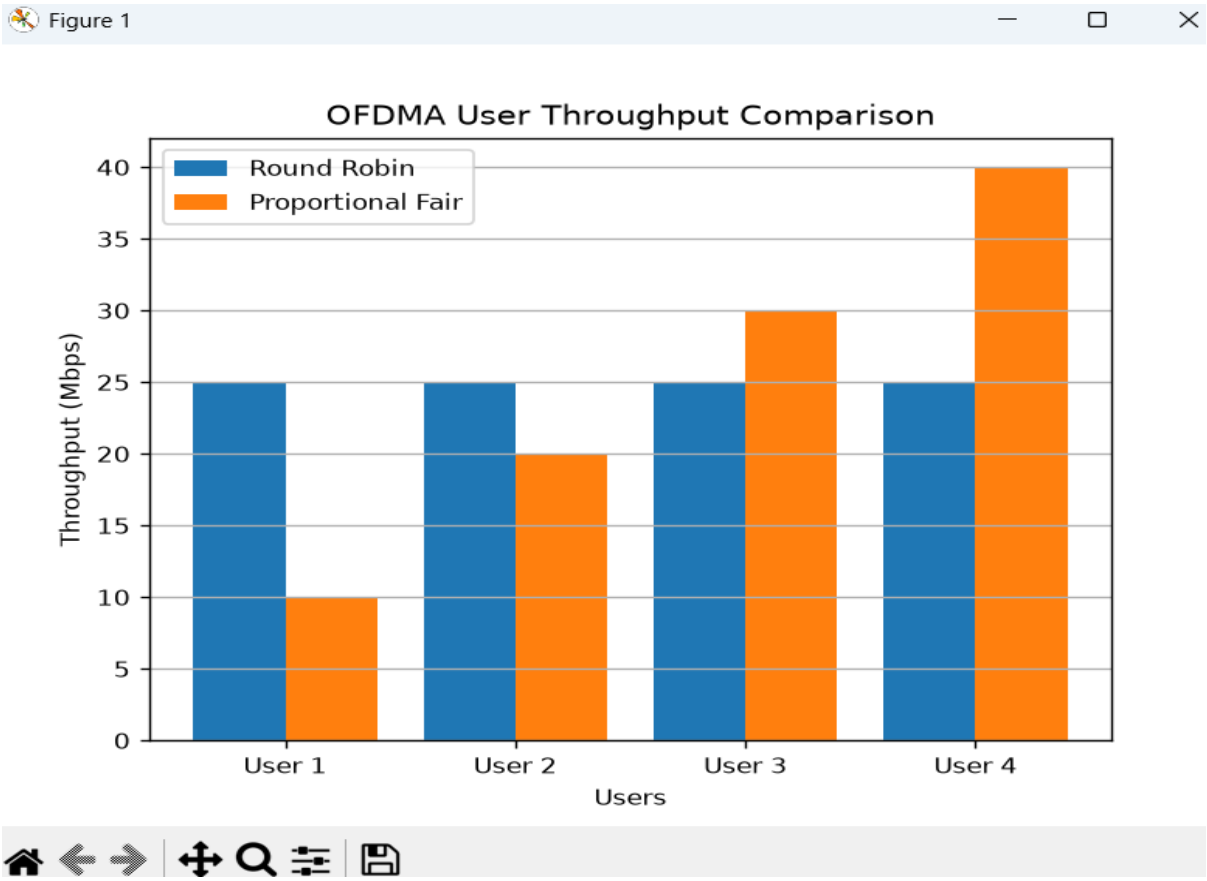
plt.bar(x - 0.2, rr, 0.4, label="Round Robin")
```

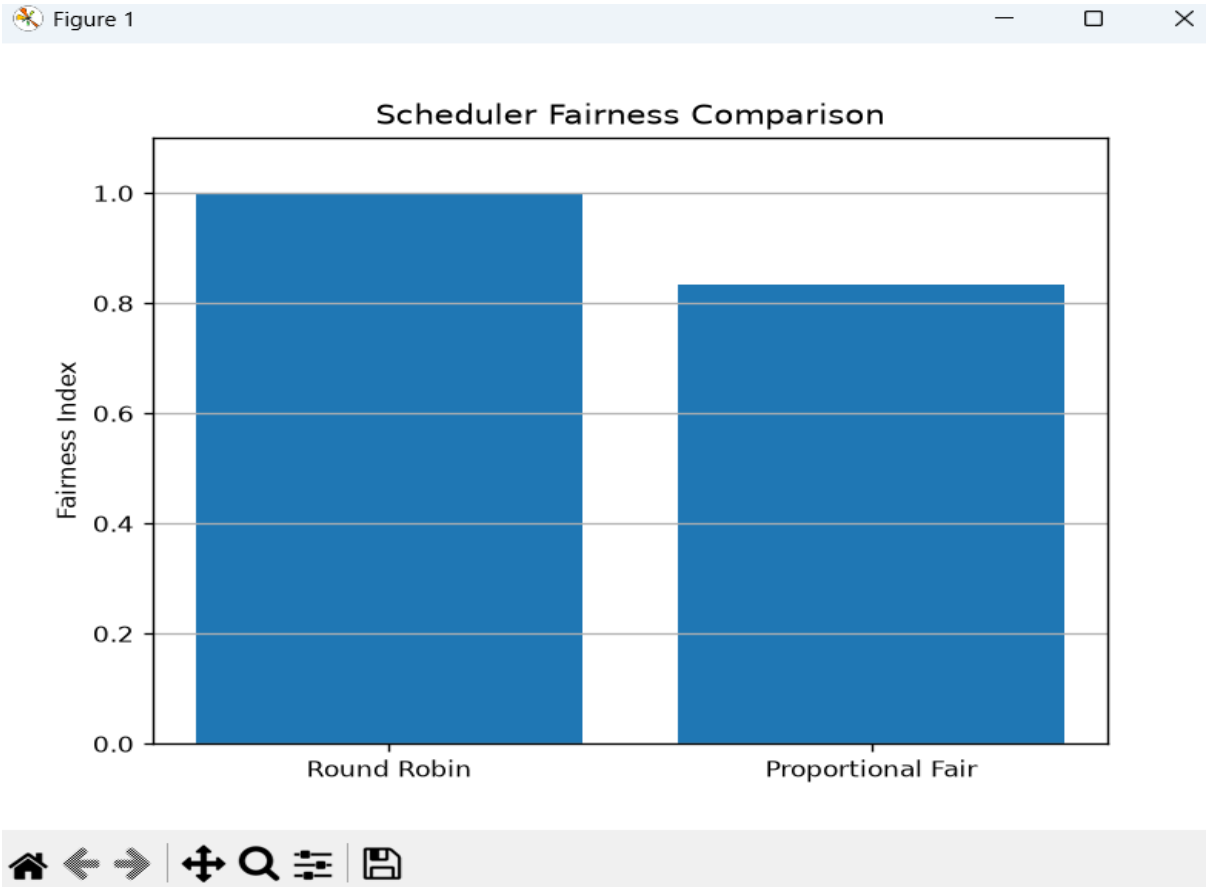
```
plt.bar(x + 0.2, pf, 0.4, label="Proportional Fair")
plt.xlabel("Users")
plt.ylabel("Throughput (Mbps)")
plt.title("OFDMA User Throughput Comparison")
plt.xticks(x, ["User 1", "User 2", "User 3", "User 4"])
plt.legend()
plt.grid(axis="y")
plt.show()
# Fairness graph
plt.bar(["Round Robin", "Proportional Fair"],
        [fairness(rr), fairness(pf)])
plt.ylabel("Fairness Index")
plt.title("Scheduler Fairness Comparison")
plt.ylim(0, 1.1)
plt.grid(axis="y")
plt.show()
```

```
======== RESTART: D:/knowledge_Engineering/mobile computing/program06.
----- OFDMA Scheduling Results -----
User    RR Throughput    PF Throughput
User 1      25.00 Mbps       10.00 Mbps
User 2      25.00 Mbps       20.00 Mbps
User 3      25.00 Mbps       30.00 Mbps
User 4      25.00 Mbps       40.00 Mbps

Round Robin Fairness: 1.0
Proportional Fairness: 0.83

Result: User throughput and fairness calculated successfully.
>>>
```


Figure 1
OFDMA User Throughput Comparison
Round Robin
Proportional Fair
Throughput (Mbps)
0
5
10
15
20
25
30
35
40
User 1
User 2
User 3
User 4
Users


Figure 1
Scheduler Fairness Comparison
Fairness Index
0.0
0.2
0.4
0.6
0.8
1.0
Round Robin
Proportional Fair